

Earth SIRRUSH: A Russia-Aligned Intrusion Set With 4 Years of Evolving Espionage Tooling

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Key takeaways

- Earth SIRRUSH (overlapping with CERT-UA's UAC-0099) has conducted spear-phishing campaigns against Ukrainian government, defense, border guard, and logistics targets since at least 2022.
- The intrusion set has developed more than 10 malware families, ranging from ones built in PowerShell and Golang to those built in .NET Framework, including a previously undocumented .NET infostealer and RAT family we named ASHVEIN, internally called "TelemetryBrowser" by its developers.
- Cross-family continuity is supported by several independent indicators, including a reused PKCS #7 signature file, structurally identical code blocks, recurring build-environment artifacts, and shared operational conventions.
- Most of the confirmed C&C domains use a single registrar (Regery.com), Cloudflare fronting, and backend IPs concentrated in AS399629 (BL Networks).

Since the start of the Russian invasion of Ukraine in 2022, Russia-aligned threat actors have conducted sustained cyber campaigns against Ukrainian government, defense, border guard, and logistics organizations. TrendAITM Research recently described how Earth Dahu and SHADOW-EARTH-066 continued to exploit an old WinRAR flaw to attack Ukrainian targets, and also reported that Pawn Storm (APT28) targeted Ukrainian and other government entities using the PRISMEX backdoor.^{1, 2} The intrusion set examined here, Earth SIRRUSH, was previously tracked as SHADOW-EARTH-065.

Activity overlapping with Earth SIRRUSH was first documented by the Computer Emergency Response Team of Ukraine (CERT-UA) as UAC-0099 in June 2023 and subsequently analyzed by Deep Instinct in December 2023.^{3, 4} Active since at least mid-2022, the group has run spear-phishing campaigns against Ukrainian government agencies, defense forces, and border guard units, with each wave of lures tuned to the institution it impersonates. Over four years of operations, it has moved from PowerShell- and Go-based tools, to compiled C#, and now to .NET Reactor-protected binaries hidden inside steganographic image files.

This report consolidates Earth SIRRUSH activity observed from 2022 through July 2026 to trace continuity across successive toolsets. Previously reported campaigns are summarized for context, while the detailed analysis focuses on newly identified malware, infrastructure, telemetry, and cross-campaign links.

1. Tooling evolution

Over four years of activity, Earth Sirrush has repeatedly replaced its toolset while keeping the same focus on Ukrainian government, defense, border guard, and logistics targets. Table 1 summarizes the campaigns and malware families in chronological order, from early PowerShell- and Go-based tooling to the latest .NET loaders and steganography-delivered payloads. Earlier campaigns already reported by other security researchers have been included as context for our own analysis.

Period	Tooling/Campaign	Public context	Contribution to this report
2022 – 2024	LONEPAGE; THUMBCHOP; CLOGFLAG; SEAGLOW; OVERJAM	Previously reported by CERT-UA and Deep Instinct	Chronological baseline for later linkage
2024 – 2025	MATCHBOIL; MATCHWOK; DRAGSTARE	Previously reported by CERT-UA; DRAGSTARE also analyzed by FortiGuard Labs	Code and build-environment comparison
October 2025	ASHVEIN; AnswerFromPolice delivery component	Internally named TelemetryBrowser by its developers	New malware, delivery-chain, and cross-family linkage analysis
February – April 2026	CINDERBLOT / BadPaw; MeowMeow; KittyCat / CyberFly delivery	BadPaw and MeowMeow reported by ClearSky	Additional CINDERBLOT delivery waves, campaign artifacts, and linkage analysis
April – July 2026	LUNCHPOKE; BURNYBEAR; MATCHBOIL.V2	Previously reported by CERT-UA	Configuration, infrastructure, and steganography linkage

Table 1. Campaign and malware overview

2. Early operations (2022 – 2024): LONEPAGE family

The earliest Earth Sirrush tooling included the PowerShell loader LONEPAGE, the C# browser stealer THUMBCHOP, and two Go-based backdoors, SEAGLOW for interactive access and OVERJAM for reverse-proxy tunnels. These tools were delivered through WinRAR SFX archives, HTA files, and CVE-2023-38831 exploitation, with command and control (C&C) conducted over plain HTTP on high-numbered ports.⁵ CERT-UA first documented this activity in June 2023, and Deep Instinct published additional analysis in December 2023 covering the WinRAR exploitation and LONEPAGE C&C protocol.

3. MATCHBOIL and ASHVEIN (2024 – 2025)

By 2025, the intrusion set had overhauled its toolkit. CERT-UA documented MATCHBOIL (a C# loader), MATCHWOK (a C# backdoor with AES-256 encrypted tasking), and DRAGSTARE (a C# stealer with DPAPI credential theft and anti-VM checks), followed by a report from FortiGuard Labs.^{6, 7} Through infrastructure pivoting from CINDERBLOT's C&C domain, we identified a related .NET malware family that we named ASHVEIN, internally called "TelemetryBrowser" by its developers. ASHVEIN is an infostealer and remote access trojan (RAT) targeting Ukrainian government personnel.

Five builds were compiled between Oct. 8 and Oct. 23, 2025, across three distinct packing variants. ASHVEIN overlaps functionally with DRAGSTARE in credential theft, screenshots, file collection, and WMI fingerprinting, but key differences separate them. DRAGSTARE was compiled by the NordDragon developer account, targets both Chrome and Firefox, and includes anti-VM checks and subnet scanning. ASHVEIN, compiled by the dev account, uses a different packing approach. The functional overlap, combined with separate build environments, indicates parallel tool development under different developer accounts for the same operational requirement. ASHVEIN's core functionality includes:

- DPAPI-based credential theft from both Chrome and Firefox browsers.
- GDI-based screenshot capture.
- File enumeration and retrieval.
- PowerShell remote shell execution.
- System fingerprinting through five WMI queries (Win32_BIOS, Win32_ComputerSystem, Win32_DiskDrive, Win32_BaseBoard, Win32_Processor).
- C&C protocol using `System.Net.Http.HttpClient` with `async/await`, TLS with certificate validation bypassed via `RemoteCertificateValidationCallback`, and a custom `TelemetryBrowserContext` User-Agent string.
- Tasking data embedded inside invisible HTML elements: The server returns pages containing `<footer style="display:none">` elements, and ASHVEIN extracts the payload with a regex (Figure 1).
- Encryption using AES with PBKDF2-derived keys and RSA for key exchange.
- A fallback C&C resolution mechanism, added by some variants, using a GitHub-based dead-drop resolver at `raw.githubusercontent.com/jsox-hub/RedirectDsa335643ededddlink/.../url.txt` (the repository has since been deleted).
- Monitoring for 15 security analysis tools including Wireshark, IDA, OllyDbg, Fiddler, and Process Monitor (Figure 2).

```

using System;
using System.Text.RegularExpressions;

// Token: 0x0200000B RID: 11
public class GClass7
{
    // Token: 0x0600002F RID: 47 RVA: 0x00002E50 File Offset: 0x00001050
    public static string smethod_0(string string_0)
    {
        string pattern = "]]veff2footer[^>]*style=\\\"display:\\\"s*none\\\"[^>]*>(.*)</footer>" /* Class8.smethod_0(..., 6) */;
        bool flag = false;
        Match match;
        while (!flag)
        {
            match = Regex.Match(string_0, pattern, RegexOptions.IgnoreCase | RegexOptions.Singleline);
            flag = true;
        }
        if (match.Success)
        {
            return match.Groups[1].Value.Trim().Trim();
        }
        return "";
    }
}

```

Figure 1. ASHVEIN Build B HTML dead-drop extraction via hidden footer element
(SHA-256: 288de8548bc4949080c7956c5698924d8e1bf9e9eff635b1a4e1cdeb088e73a4)

```

public bool method_0()
{
    List<string> source = this.method_1();
    using (List<string>.Enumerator enumerator = this.method_2().GetEnumerator())
    {
        while (enumerator.MoveNext())
        {
            GClass9.Class7 @class = new GClass9.Class7();
            @class.string_0 = enumerator.Current;
            try
            {
                if (source.Where(new Func<string, bool>(@class.method_0)).Any<string>())
                {
                    return true;
                }
            }
            catch
            {
                bool flag = false;
                bool result;
                while (!flag)
                {
                    result = false;
                    flag = true;
                }
                return result;
            }
        }
    }
    return false;
}
// [...]

// --- GClass9.cs lines 57-73: tool name list ---
// Token: 0x06000038 RID: 56 RVA: 0x00003034 File Offset: 0x00001234
private List<string> method_2()
{
    string text = "wireshark tshark dumpcap idaq ollydbg proxifier procmon netmon GlassWire Fiddler charles procexp nmapve9b8 PRTG Postman" /*
    List<string> list = new List<string>();
    foreach (string text2 in text.Split(new char[]
    {
        . .
    })))
    {
        if (text2.Length > 3)
        {
            list.Add(text2.ToLower().Trim());
        }
    }
    return list;
}

```

Figure 2. ASHVEIN Build B anti-analysis tool detection list
(SHA-256: 288de8548bc4949080c7956c5698924d8e1bf9e9eff635b1a4e1cdeb088e73a4)

There are several delivery methods for ASHVEIN, including DLL sideloading (using FORGECLAMP), VHD container, and a dedicated .NET dropper. AnswerFromPolice is a .NET executable purpose-built for ASHVEIN delivery. It embeds a DOCX decoy document as a .NET resource named AnswerFromPolice.Resources.Vidpovid.docx. The DOCX shown in Figure 3 purports to be a response from the National Police of Ukraine. While the decoy is displayed, the dropper downloads TFU_up.bin from [hxxps://shareitsimple\[.\]com/download.php?file=TFU_up.bin](https://shareitsimple[.]com/download.php?file=TFU_up.bin) and saves it to C:\Users\Public\Documents\TFU_up.exe. The dropper was distributed through a GitHub repository (bohemiaUkraine/Answers) and tinyurl[.]com redirects.

An OPSEC detail worth noting: The embedded DOCX's internal metadata (Application field) indicates that it was authored in LibreOffice 7.3.7.2 on Linux x86_64, although metadata can be manipulated. AnswerFromPolice's Program Database (PDB) path has dev in its username part, the same account used by all ASHVEIN builds.

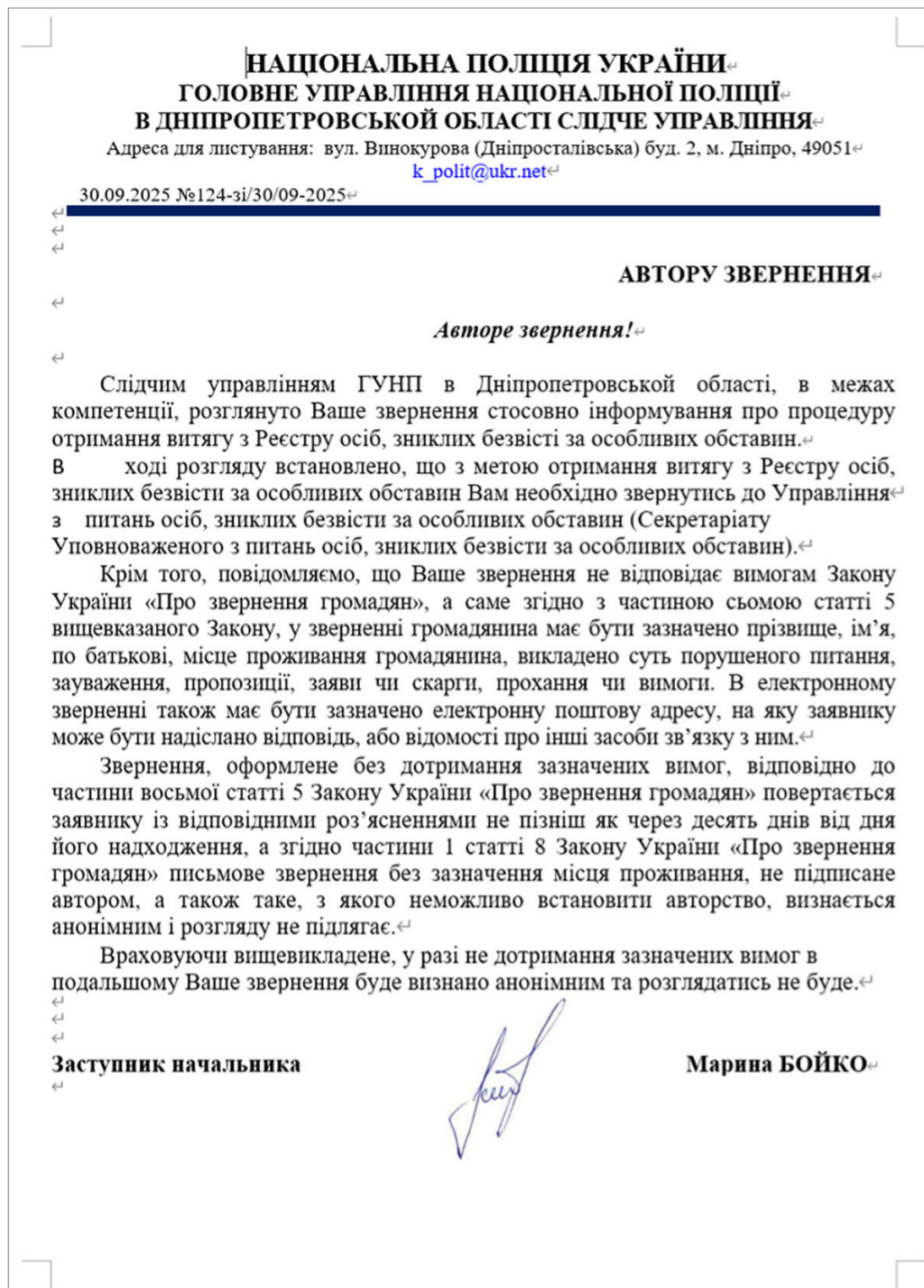


Figure 3. AnswerFromPolice embedded decoy document impersonating the National Police of Ukraine's branch in Dnipropetrovsk Oblast (SHA-256: 192e53a8879efe59ddbd4597ed64b25fb95bdfb21100de235a2423386ffaf5e)

4. CINDERBLOT and MeowMeow (2026)

In February 2026, a spear-phishing campaign impersonating the State Border Guard Service of Ukraine (DPSU) was identified, as shown in Figure 4. This campaign was first publicly reported by ClearSky Cyber Security in March 2026 under the names CINDERBLOT (BadPaw) and MeowMeow.⁸

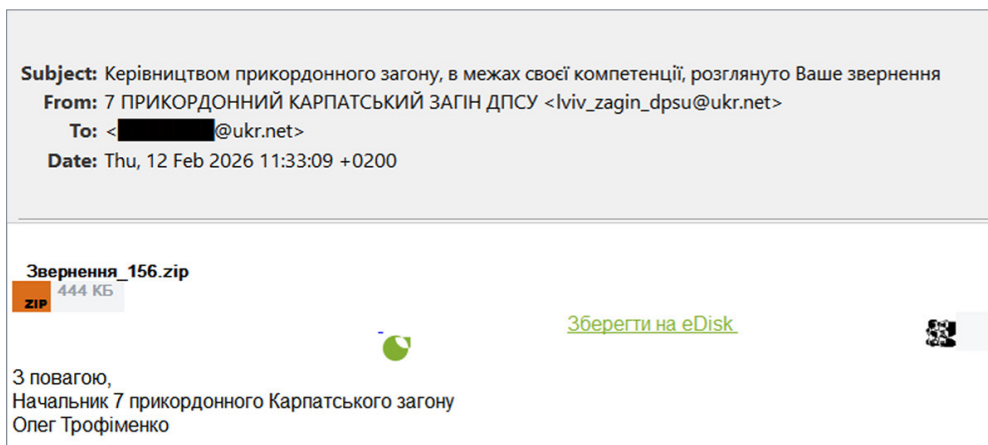


Figure 4. CINDERBLOT spear-phishing email impersonating DPSU, dated February 2026

5. April 2026 campaigns: CINDERBLOT via Drone Lure

A phishing email sent on April 20, 2026, from info[@]cyberflysystems[.]com used a procurement order for drone or unmanned aerial vehicle (UAV) parts as a lure, as shown in Figure 5. The email contained a cutt[.]ly shortened link redirecting to order.cyberflysystems[.]com (Figure 6), a professionally designed phishing site that displays a fake “ПЕРЕВІРЕНО АНТИВІРУСОМ” (“Verified by antivirus”) badge above a download button serving weaponized ZIP archives. The cyberflysystems[.]com (Figure 7) domain uses Cloudflare DNS but a different nameserver cluster from the main C&C domains.

The ZIP archive from order.cyberflysystems[.]com contains a VBS dropper. Its execution flow closely mirrors the infection chain observed in the February campaign and ultimately results in CINDERBLOT infection:

- Uses a steganographic PNG named KittyCat.png, echoing the February campaign’s CAT.png naming convention, delivered through filemail[.]com subdomains
- Checks whether the operating system was installed within the previous 10 days (DAYS_LIMIT=10) to identify and avoid potential sandbox environments
- Downloads DOCX decoy and steganographic PNG (KittyCat.png) from filemail[.]com subdomains
- Extracts embedded Res.zip from PNG via <STEGO_START> marker to C:\Users\Public\Libraries\Resources\, which contained ResourceExplorer.exe (loader), HelpersLibraries.dll (CINDERBLOT), and Microsoft.Win32.TaskScheduler.dll (legitimate DLL)
- Persists ResourceExplorer.exe via schtasks.exe LOLBin copy (C:\Users\Public\Documents\taskcreator.exe), scheduled task Helpers\Resources running every 5 minutes with finderstart argument

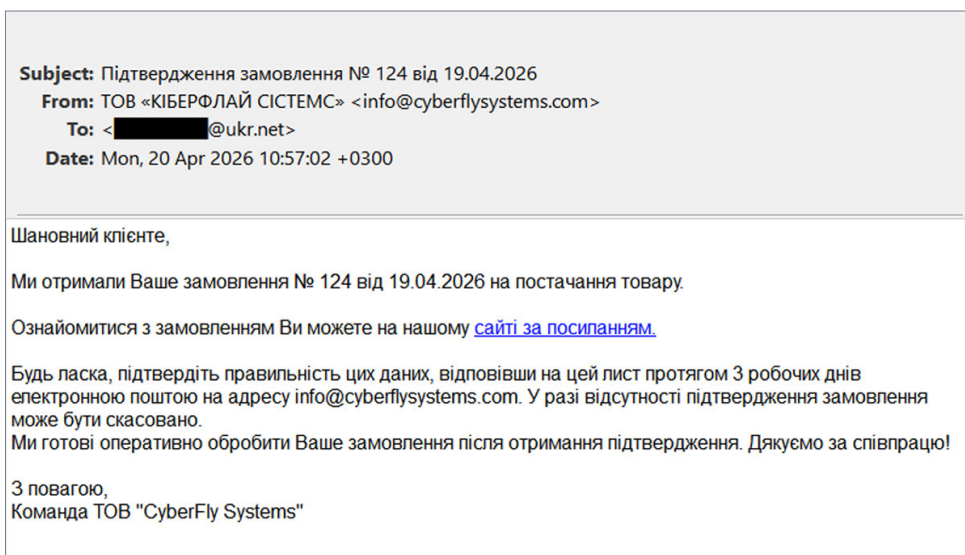


Figure 5. Spear-phishing email from the fake drone company cyberflysystems[.]com

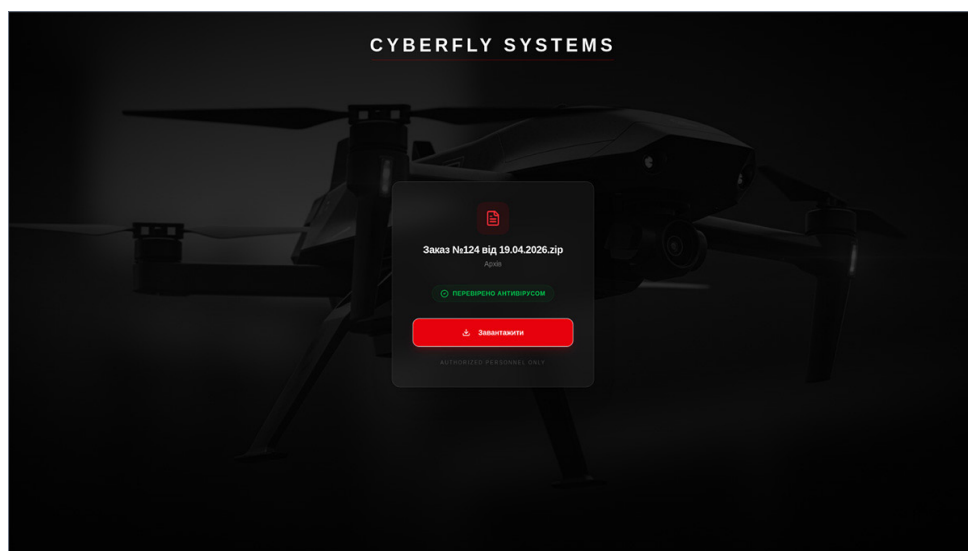


Figure 6. The order.cyberflysystems[.]com download page with a fake antivirus badge

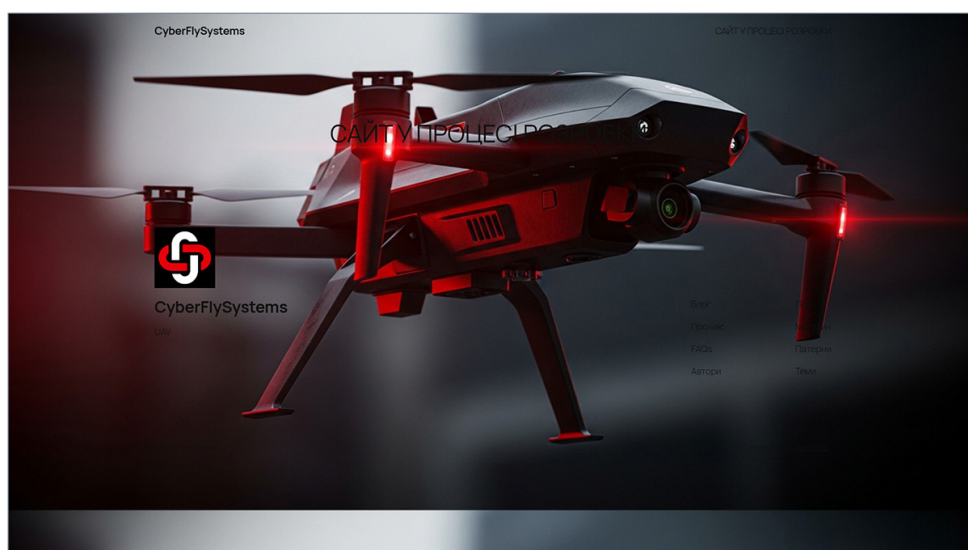


Figure 7. The cyberflysystems[.]com homepage, a fake Ukrainian drone or UAV company

6. CERT-UA July 2026 advisory: LUNCHPOKE, BURNYBEAR, and MATCHBOIL.V2

On July 21, 2026, CERT-UA published an advisory documenting a tactical shift by UAC-0099, overlapping with activity that our tracking for Earth Sirrush places between April and July 2026.⁹ The advisory introduces three new tool names: LUNCHPOKE, a 64-bit .NET DLL that masquerades as a Notepad++ plugin (NppExport.dll); BURNYBEAR, a .NET loader that downloads and executes payloads; and MATCHBOIL.V2, an updated version of the MATCHBOIL loader with AES-CBC encryption replacing the original XOR string obfuscation.

The analysis of LUNCHPOKE revealed a DLL proxy mechanism forwarding all six standard NppExport plugin exports to original-NppExport.dll while injecting malicious behavior through DllMain. The kill chain comprises the following steps:

1. The kill chain begins with a phishing email containing an image link that redirects through cutt[.]ly to a file-hosting service (filemail[.]com, easysend[.]co, or cdn.imageurlgenerator[.]com).
2. The downloaded ZIP contains a VBS dropper that uses the same U+205F Unicode padding technique observed in CINDERBLOT delivery packages.
3. The VBS downloads a decoy document and a trojanized Notepad++ 8.8.3 installation (distributed as Evernote.zip).
4. When Notepad++ launches, it loads the LUNCHPOKE DLL from its plugins directory, which downloads a password-protected RAR archive (password: UPDATERclient), extracts BURNYBEAR and MATCHBOIL.V2, and deploys them to randomized subfolders under C:\Users\Public\Libraries\. LUNCHPOKE also copies schtasks.exe to C:\Users\Public\Wallpapers\Background.exe and creates a scheduled task that runs BURNYBEAR every 3 minutes.
5. The VBS dropper checks the OS locale for Russian (1049) and Ukrainian (1058) language codes, displaying localized notification messages.

A parallel delivery track uses PowerShell steganography downloaders (NAXNh061Sr9R.ps1) hosted on cdn.imageurlgenerator[.]com. These scripts download PNG images, extract embedded ZIP archives from pixel data, and deploy executables (SearchExplorer.exe, LibUpdater.exe, ResourceExplorer.exe) with registry Run key persistence. This pixel-data extraction mechanism differs from the G0_S chunk-based steganography used in CINDERBLOT delivery, indicating the threat actor maintains multiple steganographic techniques in parallel.

Decoy documents reference Ukrainian government themes including administrative orders and judicial requests. The C&C domain is kaufen-qpon[.]icu, which resolves to 64[.]94[.]84[.]204 in the same /23 subnet as the confirmed CINDERBLOT/ASHVEIN backend at 64[.]94[.]84[.]99.

Code analysis of MATCHBOIL.V2 revealed three distinct configurations (Figure 8). Waves 1 and 2 both use kaufen-qpon[.]icu as their C&C domain, but they carry different authentication tokens and download WinRAR from different sources (cdn.imageurlgenerator[.]com for Wave 1, Dropbox for Wave 2). A third, steganography-delivered variant used a different RAR password (SMTPclient versus FILEclient) and a different mutex (Global\ExFind versus Global\RemoteLibInstall). The existence of a steganography-delivered MATCHBOIL.V2 variant connects the July 2026 toolset back to the steganographic delivery mechanism used in earlier campaigns.

```
string      c2Domain    = "https://kaufen-qpon.icu";
string      apiEndpoint = "/manager/";
string      authToken   = "PJ7RM7vVyR714oPYBD1PTDP15";
string      rarPassword = "FILEclient";
string      mutex        = "Global\\RemoteLibInstall";
int         interval    = 7; // minutes
int         duration    = 3650; // days (10 years)
```

Figure 8. MATCHBOIL.V2 C&C config

(SHA-256: 78cb23c739d24513b25f477183566c9fce0b329696e05482065f1bf01629c606)

In April 2026, prior to CERT-UA's July advisory, we discovered a related variant, STEELPANE, also abusing a trojanized NppExport.dll. Unlike LUNCHPOKE, STEELPANE is a Rust-compiled native x64 reflective DLL loader (export DLL name: ReflectiveInstaller.dll) and is decrypted using ChaCha20 rather than the XOR or AES-CBC schemes seen in BURNYBEAR and MATCHBOIL.V2.

7. Cross-family linkage

Since at least 2022, these operations have spanned multiple toolset rebuilds and a complete overhaul of every component between phases. Yet the developers' forensic traces persist across every iteration, providing multiple independent lines of evidence that tie all observed phases to a single intrusion set.

7.1 Shared P7S signature file

The same PKCS #7 (P7S) signature file appears in both CINDERBLOT delivery archives (2026) and UAC-0099's MATCHBOIL loader archives (2024 – 2025) documented by CERT-UA. This artifact is a legitimate Ukrainian qualified electronic signature originally created in April 2024 for an unrelated document. It serves no functional purpose in the delivery chain and is included purely as a social engineering prop, but its reuse across campaigns creates one of the cross-campaign links.

CERT-UA independently documented the same file in MATCHBOIL archives from 2024 to 2025. Its presence in both the MATCHBOIL and CINDERBLOT archives strengthens the case that the campaigns might have been conducted by the same operator.

7.2 Shared PDB developer accounts

PDB paths reveal a progression of build environments: C:\Users\user\ in early operations (2022 – 2024), C:\Users\NordDragon\ in DRAGSTARE, and C:\Users\dev\ in both ASHVEIN and the AnswerFromPolice. The dev account bridges the later toolsets, tying ASHVEIN- and AnswerFromPolice-related tools to the same development environment. The ConfuserEx-packed sideloading DLLs use a separate build environment under C:\Users\Administrator\Desktop\Debug\Canva_Project\.

Whether these accounts represent distinct individuals or the same developer working on different machines at different times cannot be determined from PDB paths alone, but the account-level linkage between ASHVEIN and its droppers indicates that tools were compiled in a shared build environment. Note that account names such as user, dev, and Administrator are individually common on Windows systems; their significance comes from the consistent reuse pattern across attributed samples rather than from the names themselves.

Family	Developer account	PDB path
THUMBCHOP (early operations stealer)	user	c:\Users\user\Documents\Visual Studio 2012\Projects\WindowsNotificationsServ\...\WindowsNotificationsServ.pdb
DRAGSTARE	NordDragon	C:\Users\NordDragon\Documents\visual studio 2012\Projects\svchost\svchost\bin\Release\obfuscated\svchost.pdb
ASHVEIN	dev	C:\Users\dev\source\repos\TelemetryBrowserContext\...\obfuscated\TelemetryBrowser.pdb
AnswerFromPolice	dev	C:\Users\dev\source\repos\AnswerFromPolice\...\AnswerFromPolice.pdb
DLL sideloading	Administrator	C:\Users\Administrator\Desktop\Debug\Canva_Project\...

Table 2. PDB path comparison

7.3 Russian-language indicators

CINDERBLOT’s decrypted configuration contains a Russian uptime regex (работоспособного\состояния\s(\d+)\сек). This regex parses the output of the systeminfo command, which returns localized strings depending on the OS language. Including the Russian variant indicates that the developer specifically supports Russian-locale Windows environments. According to ClearSky, MeowMeow contains Russian military-technical debug strings, including the phrase “Время выхода на рабочий режим” (“Time to reach operational readiness”). These are indicators that might be consistent with Russian-speaking developers, although the possibility of deliberate false-flag indicators cannot be excluded.

7.4 Shared XOR cipher family

ASHVEIN, MATCHBOIL, and DRAGSTARE use an identical byte-swap XOR string encryption algorithm with per-variant seed constants (227668699, 938305803, 680262394), indicating shared codebase across these three families (Figure 9). While each family uses different assembly GUIDs and packing frameworks (Sugar, ConfuserEx, or no packing), the underlying XOR byte-swap string cipher is structurally identical across all three, with only the seed constant varying. ASHVEIN Build C/D uses a variant seed 941499699 with the same algorithm structure.

```

internal static string smethod_0(string string_0, int int_0)
{
    int num = 227668699 + int_0;
    char[] array = string_0.ToCharArray();
    for (int i = 0; i < array.Length; i++)
    {
        char[] array2 = array;
        int num2 = i;
        char c = array[i];
        array2[num2] = (ushort)((((int)(c & 'y') ^ num++) << 8 | (int)((byte)((int)(c >> 8) ^ num++)));
    }
    return string.Intern(new string(array));
}

```

Figure 9. Shared XOR cipher across ASHVEIN, MATCHBOIL, and DRAGSTARE
(SHA-256: 9c68e5d111a581a195284f884d29c5576723264a01d80a20729e09510fb6078b)

7.5 Identical WMI fingerprinting queries

Both CINDERBLOT and ASHVEIN implement system fingerprinting through five WMI queries that are textually identical, including letter case and SQL formatting:

```

SELECT SerialNumber FROM Win32_BIOS
SELECT SerialNumber FROM Win32_DiskDrive WHERE Index=0
SELECT SerialNumber FROM Win32_BaseBoard
SELECT ProcessorId FROM Win32_Processor
Select * from Win32_ComputerSystem

```

The identical queries, including the anomalous lowercase Select on the final query versus uppercase SELECT on the others, are consistent with a shared codebase rather than independent development. This fingerprinting block appears in MATCHBOIL/ASHVEIN (ASHVEIN, DRAGSTARE) and CINDERBLOT, linking both phases at the code level.

7.6 ‘TelemetryUP’ branding crossover

ASHVEIN’s Portable Executable (PE) version info CompanyName field is set to “TelemetryUP” across all builds (Figure 10). The ASHVEIN C&C domain crazy-love[.]app hosts a landing page at /documentation/uptelem.html titled “Telemetry UP! Soft for block Telemetry”, a facade page built from a tooplate[.]com template with a fake “Provided by Google” footer (Figure 11). The shared branding between malware metadata and C&C landing page indicates that ASHVEIN and its associated C&C infrastructure are maintained by the same development or operational entity.

```

using System;
using System.Diagnostics;
using System.Reflection;
using System.Runtime.CompilerServices;
using System.Runtime.InteropServices;
using System.Runtime.Versioning;

[assembly: AssemblyVersion("1.1.2.0")]
[assembly: AssemblyTitle("Telemetry Context")]
[assembly: AssemblyDescription("")]
[assembly: AssemblyConfiguration("")]
[assembly: AssemblyCompany("TelemetryUP")]
[assembly: AssemblyProduct("")]
[assembly: AssemblyCopyright("Copyright © 2025")]
[assembly: AssemblyTrademark("TLCK")]
[assembly: ComVisible(false)]
[assembly: Guid("a3cabfce-218e-4640-bd9e-923e7943b1e1")]
[assembly: AssemblyFileVersion("1.1.2.0")]

```

Figure 10. ASHVEIN Build B AssemblyInfo.cs: CompanyName = TelemetryUP
(SHA-256: 288de8548bc4949080c7956c5698924d8e1bf9e9eff635b1a4e1cdeb088e73a4)



Figure 11. The crazy-love[.]app/documentation/up telem.html landing page

7.7 Shared code conventions

Both CINDERBLOT and ASHVEIN stage payloads in C:\Users\Public\Libraries\, store C&C configuration in .library-ms files (an uncommon file extension that blends with Windows Libraries), and implement the same hybrid encryption scheme: AES with PBKDF2-derived keys (Rfc2898DeriveBytes) and RSA for key exchange. Both use the same TLS certificate bypass via RemoteCertificateValidationCallback, the same Debugger.IsAttached anti-debug check, and C&C protocols that hide tasking inside semantically invisible HTML elements. Both target .NET Framework 4.6.

8. Infrastructure

Figure 12 summarizes the later-phase infrastructure assessed as actor-controlled. Confirmed later-phase C&C domains in this cluster were registered through Regery.com and fronted by Cloudflare. Separate delivery infrastructure, including shareitsimple[.]com, falls outside this pattern. The backend IPs behind the Cloudflare fronting, identified via historical DNS records predating the shift to Cloudflare, cluster in AS399629 (BL Networks),¹⁰ an autonomous system also used by unrelated actors; ASN (autonomous system number) co-location therefore supports linkage only when combined with other evidence.

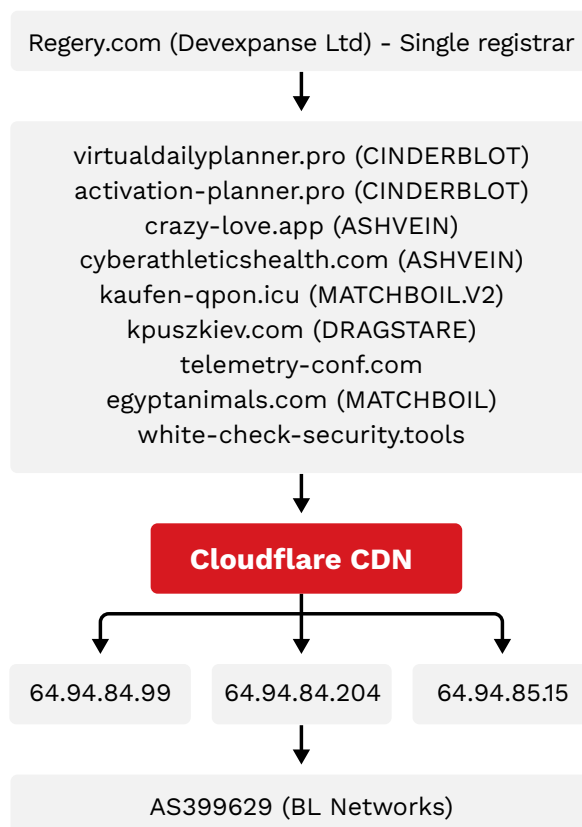


Figure 12. Earth SIRRUSH C&C infrastructure: single-registrar domain procurement, Cloudflare fronting, and backend IP clustering

8.1 Shared backend IP

The backend IP 64.[.]94.[.]84.[.]99 directly links CINDERBLOT (virtualdailyplanner[.]pro) and ASHVEIN (crazy-love[.]app) at the server level, providing infrastructure-based evidence connecting MATCHBOIL/ASHVEIN and CINDERBLOT tools. A third domain, white-check-security[.]tools, also resolved to this IP.

8.2 Cross-cluster IP bridge

Two backend IPs (185.[.]53.[.]46.[.]190 and 85.[.]209.[.]195.[.]161) appear in the DNS history of both cyberathleticshealth[.]com and telemetry-conf[.]com, domains that otherwise appear to be managed separately behind Cloudflare. The shared backend IPs confirm that the separation is operational compartmentalization, not distinct operators.

8.3 Registrar pattern

Every known MATCHBOIL/ASHVEIN and CINDERBLOT domain was registered through Regery.com (Devexpanse Ltd). This single-registrar pattern, spanning domains used by two malware families across seven months of registration activity, provides additional infrastructure-level evidence of a single operator. Separate delivery infrastructure was used for payload hosting: infotrackerstatistic[.]live for email tracking and ZIP delivery in the CINDERBLOT campaign, and shareitsimple[.]com as a non-Cloudflare payload hosting server for the AnswerFromPolice dropper. Both sit outside the main C&C domain cluster.

9. Victimology

Telemetry data and spear-phishing lure themes indicate targeting across:

- Government agencies and ministries.
- Defense forces and border guard units (DPSU).
- Transport and logistics operators.
- Defense supply chain (drone or UAV procurement).

At a transport and logistics operator, the implant maintained persistent access over an extended period, with C&C beaconing surviving a month-long offline gap through scheduled-task persistence.

Lure themes reference Ukrainian government institutions including the National Police, the State Border Guard Service, tax authorities, and the Ministry of Justice.

10. Attribution

Using our attribution framework,¹¹ we assess with high confidence that the activity described here belongs to a single tracked intrusion set, Earth SIRRUSH, which overlaps with CERT-UA's UAC-0099. The assessment is based on the combined weight of code-level similarities, reused artifacts, infrastructure relationships, and consistent targeting.

No single indicator, including common PDB usernames, targeting patterns, or shared backend systems, is sufficient on its own. Their convergence supports the cluster-level assessment. ESET has separately assessed that UAC-0099 might conduct initial-access operations for Sandworm (APT44), an intrusion set linked to Russia's military intelligence agency and known for destructive operations against Ukraine.¹²

11. Risk management guidance

Based on the observed tactics, techniques, and infrastructure, organizations targeted by Earth SIRRUSH should apply layered controls to detect the intrusion set's phishing, steganographic delivery, persistence, and C&C activity.

11.1 Network-level controls

- Block all C&C domains and backend IPs listed in the "Indicators of compromise (IoCs)" subsection of the "Appendix" section of this report at the DNS and firewall level.
- Alert on outbound connections to addresses within AS399629 (BL Networks). Note that this ASN is not exclusively used by this intrusion set; use in conjunction with other indicators to reduce false positives.

11.2 Endpoint detection

- Monitor for executable file creation in:
 - C:\Users\Public\Libraries\ and subdirectories.
 - %LOCALAPPDATA%\UpdateHelpers\.
- Detect schtasks.exe copied to non-standard locations (for example, C:\Users\Public\Documents\taskcreator.exe for CINDERBLOT, C:\Users\Public\Wallpapers\Background.exe for LUNCHPOKE)
- Watch for mshta.exe execution followed by file writes to user-writable staging directories.
- Monitor for Notepad++ loading DLLs from nonstandard plugin directories.

11.3 File-based indicators

- Scan for PE files embedded after PNG IEND chunks (steganographic payload delivery).
- Monitor for VHD file mounts from email attachments or web downloads.

11.4 Email security

- Filter attachments with U+205F Unicode padding in filenames (extension masking technique).
- Scrutinize emails from ukr.net accounts claiming to be government institutions.

11.5 Identity and access

- Enable DPAPI audit logging to detect credential theft attempts against Chrome and Firefox browsers.
- Monitor for abnormal access to Chrome's Local State file and Firefox profiles.ini.
- Enable PowerShell ScriptBlock logging and Constrained Language Mode.

11.6 Organizational measures

- Brief personnel in logistics, border guard, and defense-adjacent sectors on spear-phishing tactics, techniques, and procedures (TTPs) targeting Ukrainian organizations.
- Validate digital signatures on incoming documents, as reused or mismatched P7S files might indicate social engineering.

12. Conclusion

Earth SIRRUSH has been active since at least 2022, with its tooling undergoing multiple rebuilds. Each toolkit is replaced by one harder to detect: from PowerShell scripts and Go binaries, to compiled C#, and now to .NET Reactor-protected implants delivered inside images. The activity documented here spans 2022 through July 2026, showing that despite those rebuilds, forensic linkage across phases indicates that swapping tools does not erase attribution.

Available evidence suggests that the targeting has expanded beyond government and military organizations to include civilian logistics and infrastructure operators that keep Ukraine supplied. That drift tracks the war: As the conflict continues, the value of understanding Ukraine’s logistics networks rises, and the cyber effort follows the same logic as the kinetic one. The implant’s built-in C&C failover and its survival across a month-long offline gap point to an operation built for the long haul, not a smash-and-grab.

13. TrendAI Vision One™ Threat Intelligence Hub

The TrendAI Vision One™ Threat Intelligence Hub provides the latest insights on emerging threats and threat actors, exclusive strategic reports from TrendAI™ Research, and the TrendAI Vision One™ Threat Intelligence Feed in the TrendAI Vision One™ platform.^{13, 14}

Threat actor: Earth SIRRUSH¹⁵

Emerging threat: Earth SIRRUSH: A Russia-Aligned Intrusion Set With 4 Years of Evolving Espionage Tooling¹⁶

TrendAI Vision One™ Intelligence Reports (IoC sweeping): Earth SIRRUSH: A Russia-Aligned Intrusion Set With 4 Years of Evolving Espionage Tooling¹⁷

TrendAI Vision One™ customers can retrieve the indicators associated with this activity for retrospective sweeping across their environments.

14. Appendix

14.1 Indicators of compromise (IoCs)

Type	Indicator of compromise (IoC)	Description
sha256	55d8b1951e1ce8b3aea07ee3a3fb9f0e8f0cd345d08096806ddad9a6691510ec	LONEPAGE
sha256	00b7e464b190dc3ee8847e214857b83180f948b5c887466f09bb91a3352b889a	LONEPAGE
sha256	f0bdcbb507aaf9af91bc69511a4873462e9b2b07c173c71d9c4c622a2754f5d5b	Stealer
sha256	986694cad425c8f566e4e12c104811d4e8b30ce6c4c4d38f919b617b1aa66b05	CVE-2023-38831
sha256	87291b918218e01cac58ea55472d809d8cdd79266c372aeb9ee593c0f4e3b77	CVE-2023-38831
sha256	06564bd863c772ca700b2f4472538fbfa52b96bdaef20c2a1daa11a2a08d61fd	MATCHBOIL
sha256	5ea8f7650485e1c92659bed861e002b5916fa5b5a72fb9a89ee2c1035d575307	MATCHBOIL
sha256	c1f5cef846d9f2a9884bbf3343c88cce8d32a31b05ed99f28fc37aff182cf5bd	MATCHBOIL
sha256	f4f6beea11f21a053d27d719dab711a482ba0e2e42d160cefd9bdad7a958b93d0	DRAGSTARE
sha256	9c68e5d111a581a195284f884d29c5576723264a01d80a20729e09510fb6078b	ASHVEIN

Type	Indicator of compromise (IoC)	Description
sha256	288de8548bc4949080c7956c5698924d8e1bf9e9eff635b1a4e1cdeb088e73a4	ASHVEIN
sha256	a07a1764b9b795d2a9fb536f47f8b33f6f970c46b4ead5273d797fcb46c4815d	ASHVEIN
sha256	187bf4f7d551ec1347abc7936e8114fcdf45a53fab51ccbab2f585bcc478e683	ASHVEIN
sha256	595dcacea79205f27ac5446b373dba43aa8c7f3469155b8ad7b6c6f43f429284	ASHVEIN
sha256	e4322c18b9406589b0c9a183a79bd6125a40fba91ac9060df1fbf5e57a7db3d1	AnswerFromPolice
sha256	e3bbf378f34850c0c773bbd5ba010ae42f4fce76c16a70b26494f0b770394d0b	AnswerFromPolice
sha256	6cad470e10c09151b5d337a082a088cfe25d697ef295e02759e1e68e8b3bbcb	CINDERBLOT
sha256	672395dc75fe4b96bc4908180a5234fb2de34416d7327c1bdef9bced82d507ae	CINDERBLOT
sha256	02aab65e4916d5dfab0fd1dd4ea6f6e7e3e8afdbbb48940e4de3fda98b4a9d8d	CINDERBLOT
sha256	cd76ced501233b57e3c308f6cc69d414404c674c0a4502319f9fa29374c9de86	Steganography
sha256	28bef809c98de7d22951052d66002cc793b6d46994404924697d70675f6efc3e	Steganography
sha256	af36804fce2137b2f9e34b01ebe3a768eb76130c665866bcd8a83ae7bd22b5f1	MATCHBOIL
sha256	fbe4b82d3271f96780704fe52eb63af6e8a8cbdf392669f442833286d1d2823d	MATCHBOIL
sha256	2a7dc81b52c9be783cdc7df4a0f80ca7412ea904e40357fa9638cfcaa46a2ec5	MATCHBOIL
sha256	619d7a6c60231e2eb7cdba7d61ae3a7ff8817d0e49c375174b7183625ef57e4e	MATCHBOIL
sha256	800a98b81a3b355cae70aa73acb155fef932e558766d004a225bb897386e3d1c	MATCHBOIL
sha256	31412060326c226e8e8fe762217519d1e9ef79779333c422e389efde773acd3c	Steganography
sha256	0ae5a7af8edcbeefc964bcdb66f38ad1e237c6815fc435906feba6969938cf03	Steganography
sha256	c5fb4fefa8a7d705d913b37c1bfbe04109ff3b534fe5f07b226023c1237405bc	Steganography
sha256	285f788ca4153994cbc4e017222b43a2c3ae9405650707cf617bf1575e90f3b9	CERT-UA July 2026
sha256	85126d45d9fcdcbcc084e2a7fdc449337a12dd50c6a25fd8903c41f5b351556f8	CERT-UA July 2026
sha256	25859c3855587309ac3e93dce531e8eca317d69568cc63b0f3c1f9466919c312	CERT-UA July 2026
sha256	87a574b6c3c0d2229c869388875c698fe491776dab7724e4cf3aded2c67b5364	CERT-UA July 2026
sha256	4453bf8cd981b35f8defe5c8392d6b23cea730877c047d99e11535b20da0caca	CERT-UA / Initial loader
sha256	2954fcfab26952f9b190fb648d7b6ed84c0e062c848e4a8b77901b5e3abc521b	ASHVEIN
sha256	d648c88e92a5532c4773c97edd19c2e8b2ab34c4b2b3cbb95caa1d05bcaa4abe	DLL sideloading
sha256	2aba06046b6f69960534de758d886959f725eee5c370a4e003a1a71069380593	CINDERBLOT
sha256	f1d8901197a2856b2656e7080b65cd0137a0e7058b8955a10381f706411dcc4b	CINDERBLOT
sha256	61f035513650b2a37007ed2dbf9059ef529e32c5945b6339f081881f5007f8b6	CINDERBLOT
sha256	642cabb0e96000baa28a8268816f2b06d8a917169b975a35f286783664b19121	CINDERBLOT
sha256	a01ce089f575ddf5376abd49fdb21ac2b2cddb35d53030f2425949bd126009fb2	CINDERBLOT
sha256	9f13b4bcc57e24bfe5d08c738a92cfca5b5fe8d8fa327226fffd813ddf1bdea1	CINDERBLOT (deobf)
sha256	fc38b9a99402360f4e8061ae2e5b4a941b401b5505f7e718ac5fc22b5a36d737	CINDERBLOT (deobf)
sha256	b99542d575b26428f00fc2859540c844156a6eb76976611d736f8f596980454e	CINDERBLOT (deobf)
sha256	047563685277326041325bf781179ce80f4deec20a2777db8caf5767ff12cbcb	CINDERBLOT (deobf)
sha256	dd662933b0024fbc1828c8fd40df3156958dad94ba154ecd5eb009510db9c021	CINDERBLOT
sha256	6770bef57afae63c935bcf9512fe8161522efa3aaff65611ad9cb2f2d34b48d7	FILEMAIL delivery
sha256	b4f2028a631d27827444f668d634d86ccd85430bf2204974213654d14f22bdd2	FILEMAIL delivery
sha256	a4ca9fff08e367fea9c19eb13893e9af6c11def87a8154278ce41c2a05516f26	FILEMAIL delivery
sha256	bc7b3c2fdd41afd25a1254cc5371c897204f05fcb8b42ac4a65a85b21495295e	Steganography
sha256	e5bd6c862e3c6f308eccf3b927f07192313476132145b15240593e5b909446aa	Steganography
sha256	5773aa6c41dc2dad571f07c496bea47d375d3a72328662701715b2841af72b56	NppExport variants
sha256	60898f3582ea771bee9de0d79e21e4f1911fdeefa379eb6c2230ce8d69efb5dd	NppExport variants
sha256	5030d1745b6f872b2e45d0039e743c33b4b9867f21543a22e5311b27d8af2a6d	NppExport variants
sha256	a403b1107cac48c97546a0f338d15cb00fe4142e119c18ce0558925cce9da176	NppExport variants
ipv4	156[.]38[.]245[.]115	C&C (multiple families)
ipv4	179[.]48[.]251[.]182	C&C (multiple families)
ipv4	217[.]12[.]218[.]107	C&C (multiple families)

Type	Indicator of compromise (IoC)	Description
ipv4	45[.]148[.]121[.]6	C&C (multiple families)
ipv4	62[.]210[.]204[.]94	C&C (multiple families)
ipv4	2[.]59[.]222[.]98	C&C (multiple families)
ipv4	147[.]78[.]46[.]40	C&C (multiple families)
ipv4	64[.]95[.]10[.]117	C&C (MATCHBOIL)
ipv4	64[.]95[.]11[.]80	C&C (MATCHBOIL)
ipv4	64[.]94[.]85[.]15	C&C (MATCHBOIL)
ipv4	64[.]95[.]13[.]203	C&C (MATCHBOIL)
ipv4	64[.]95[.]13[.]210	C&C (MATCHBOIL)
ipv4	64[.]94[.]84[.]99	C&C (ASHVEIN/ CINDERBLOT)
ipv4	64[.]95[.]13[.]21	C&C
ipv4	64[.]95[.]13[.]208	C&C
ipv4	64[.]95[.]10[.]223	C&C
ipv4	64[.]95[.]12[.]130	C&C
ipv4	95[.]111[.]249[.]30	C&C (ASHVEIN)
ipv4	185[.]53[.]46[.]190	C&C
ipv4	85[.]209[.]195[.]161	C&C
ipv4	64[.]94[.]84[.]204	MATCHBOIL.V2 C&C backend (AS399629, BL Networks, Sheridan WY)
domain	egyptanimals[.]com	C&C (MATCHBOIL)
domain	geostat[.]lat	C&C (MATCHBOIL)
domain	airarticlegenerate[.]com	C&C (MATCHBOIL)
domain	kpuszkiev[.]com	C&C (MATCHBOIL)
domain	secfileshare[.]com	HTA hosting (MATCHBOIL)
domain	startdeeplearning[.]com	C&C (MATCHBOIL)
domain	rutech[.]tech	C&C (MATCHBOIL)
domain	uztickets[.]com	C&C (MATCHBOIL)
domain	virtualtelemetryassistant[.]com	C&C (ASHVEIN)
domain	crazy-love[.]app	C&C (ASHVEIN)
domain	shareitsimple[.]com	C&C (ASHVEIN)
domain	virtualdailyplanner[.]pro	C&C (CINDERBLOT)
domain	infotrackerstatistic[.]live	C&C (CINDERBLOT)
domain	white-check-security[.]tools	C&C (CINDERBLOT)
domain	activation-planner[.]pro	C&C (CINDERBLOT)
domain	api-available-checker[.]com	C&C (CINDERBLOT)
domain	telemetry-conf[.]com	C&C
domain	api-servername-lookup[.]com	C&C (ASHVEIN)
domain	flycloud-service[.]com	C&C
domain	cyberathleticshealth[.]com	Payload hosting (ASHVEIN)
domain	cyberflysystems[.]com	Phishing email
domain	order.cyberflysystems[.]com	ZIP download hosting
domain	veetransfer[.]com	ZIP delivery (CINDERBLOT)
domain	easysend[.]co	File hosting (LUNCHPOKE)

Type	Indicator of compromise (IoC)	Description
domain	kaufen-qpon[.]icu	C&C (MATCHBOIL.V2)
email address	court.ics3312[@]ukr.net	Spear-phishing sender
email address	spd10381[@]ukr.net	Spear-phishing sender
email address	spd1401[@]ukr.net	Spear-phishing sender
email address	lviv_zagin_dpsu[@]ukr.net	Spear-phishing sender
email address	info[@]cyberflysystems.com	Spear-phishing sender
email address	donloda[@]ukr.net	Spear-phishing sender

The list of IoCs may also be downloaded via the link in the References section.¹⁸

14.2 MITRE ATT&CK mapping

Tactic	Technique	ID	Used by
Initial Access	Phishing: Spear-phishing Link	T1566.002	CINDERBLOT
Execution	System Binary Proxy Execution: Mshta	T1218.005	CINDERBLOT
Persistence	Scheduled Task/Job	T1053.005	CINDERBLOT, LUNCHPOKE, MeowMeow
Defense Evasion	Obfuscated Files: Steganography	T1027.003	CINDERBLOT (STEGO_START PNG, KittyCat.png)
Defense Evasion	Hijack Execution Flow: DLL Side-Loading	T1574.002	LUNCHPOKE, FORGECLAMP, ASHVEIN
Defense Evasion	System Checks (Sandbox Evasion)	T1497.001	CINDERBLOT (InstallDate), BURNYBEAR (args check), ASHVEIN (timing check)
Discovery	System Information Discovery	T1082	CINDERBLOT, ASHVEIN
Credential Access	Credentials from Password Stores	T1555.003	ASHVEIN
Command and Control	Encrypted Channel: Asymmetric Cryptography	T1573.002	ASHVEIN, CINDERBLOT
Exfiltration	Exfiltration Over C&C Channel	T1041	ASHVEIN, CINDERBLOT
Command and Control	Ingress Tool Transfer	T1105	LUNCHPOKE (RAR download), ASHVEIN (payload download), BURNYBEAR
Command and Control	Web Service	T1102	ASHVEIN Build B (GitHub dead-drop resolver)
Collection	Screen Capture	T1113	ASHVEIN, DRAGSTARE (GDI BitBlt)
Execution	Command and Scripting Interpreter: PowerShell	T1059.001	MeowMeow, CINDERBLOT (PowerShell remote shell)
Defense Evasion	Debugger Evasion	T1622	CINDERBLOT (Debugger. IsAttached), ASHVEIN
Defense Evasion	Deobfuscate/Decode Files or Information	T1140	CINDERBLOT (.NET Reactor), ASHVEIN (XOR cipher), MATCHBOIL.V2 (AES-CBC), STEELPANE (ChaCha20)
Defense Evasion	Match Legitimate Name or Location	T1036.005	LUNCHPOKE (NppExport.dll), BURNYBEAR (schtasks->Background.exe)

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